

PMA-3-1 · Fundamentals of Ad Tech, Data & Analytics · Fall 2026

LAB 1

Intro to Behavioral TV Data

Ratings, reach and frequency from one month of panel data

ISSUED	DUE	POINTS	SUBMIT VIA
Fri, October 2, 09:00	**Tue, October 13, 23:59 EEST**	12	GitHub repository link · `lab1.ipynb` (+ optional `.py`)

BUILDS ON	PLATFORM	AUTHORS
Lectures 1–2 — How Advertising Works; the Media & Ad Ecosystem	Google Colab / Kaggle	Jonathan Steuer · Maksym Shkarupylo (TA)

SCENARIO

You have just joined the research team of a US media agency. Before anyone lets you plan a campaign, your manager wants you to understand the raw material every TV price is built on: *people-meter panel data*. You get one month (October 2024) of respondent-level viewing of ten networks, five big (**CBS**, **NBC**, **ABC**, **FOX**, **Fox News**) and five small (**FX**, **Bravo**, **HGTV**, **Food Network**, **History**), with their schedules, the demographics of every panel member and their daily weights. A panel is only useful for what it says about the whole country, so every number you report is **projected to the US universe**.

1 Learning goals

By the end of this lab you should be able to:

- load a multi-table TV dataset, decode its attribute codes and describe it with a few honest numbers;
- use **weights** to turn panel viewing into US audiences;
- compute a telecast's **rating** and a program's **reach** for households and for persons;
- compute **frequency** for a program and read a **frequency distribution**;
- compare networks' audiences by age and gender.

2 Data

Why synthetic data

The dataset is **synthetic**, and you get it *as is*. Real respondent-level TV data (such as Nielsen's) is licensed, far too big and messy for a first lab, so we squeezed it into something small enough to get your hands on: one month, ten networks, about 4,000 households. The schedule is real (October 2024); the households and their viewing are simulated and calibrated to published US audiences, so program sizes are close to the real ones (for example, *Tracker* on CBS: about 6.5 million viewers here, 8.1 million published). Everyone gets the same five Parquet files.

Who is in the data

A **panel** is a sample of households whose TV viewing is measured electronically; each person presses their own button on a "people meter" when they start and stop watching. Every respondent has a type: the `resp_type` column in `viewings` and `daily_weights`, and a code (101–104) in `viewer_attributes`:

- `H`: the **household** itself. Its record carries the household's characteristics, and its viewing rows show when the TV was tuned to a network (whenever anyone at home watched it);
- `R`: a **respondent**, a member of the panel household;
- `L`: a **long-term visitor**, someone staying in the home for weeks (a relative, for example). They get their own id, like a member;
- `S`: a **short-term guest**, e.g. a friend over for one evening. A guest only presses a guest button and enters age and sex. Their id is the household's plus the button number (`81`, `82`, ...), so **two guest rows with the same id are not necessarily the same person**: guest 81 on Saturday and guest 81 on Sunday may be different friends.

Ids. Everyone has a `respondent_id`. Everyone in a household shares the same id beginning, the household number, and differs in the last two digits: the household record ends in `00`, its people in `01`, `02`, ..., its guests in `81`, `82`, ... For example, household `100001` has `10000100` (H), `10000101`, `10000102` and guest `10000181`. There is no separate household column: the household number is `respondent_id // 100`.

Households come and go: a household stays in a panel for about two years, so every month some leave and new ones join, and no field says when. It is always hard to tell new respondents from old ones; in real work you check when each respondent *first* (and last) appears in the panel, which is what `daily_weights` lets you do.

Weights: from panel to universe

A panel of a few thousand homes stands for all 125,800,000 US TV households and 317,277,664 Americans aged 2+. Every **broadcast day** (06:00–05:59, as in the TV industry), each household and each person in the panel gets a **weight**: how many US households (for `H`) or people (for `R` and `L`) they represent that day. The weight is the same for the whole broadcast day and can change the next one, because people join, leave or miss a day and the rest are re-balanced. A person weight of 37,200 means "this person stands for about 37,200 Americans today". Summing weights gives **projected** numbers: the sum of the weights of the people who watched is the **projected audience**, and each day the person weights add up to the universe (the household weights to the TV households), up to rounding.

`programs`: one row per telecast

Every minute of the month on each network belongs to exactly one telecast.

COLUMN	TYPE	MEANING
`telecast_id`	`int`	Unique telecast.
`network_code`	`str`	`CBS`, `NBC`, `ABC`, `FOX`, `FNC` (Fox News), `FX`, `BRAVO`, `HGTV`, `FOOD`, `HIST`.
`network_name`	`str`	Full network name, e.g. `Fox News Channel`.
`network_type`	`str`	`broadcast`, `cable_news`, `cable_ent`.

COLUMN	TYPE	MEANING
program_name	str	Program title. Hours a broadcast network leaves to its local stations are Local / Syndicated Programming.
episode_name	str	Episode title or game; empty for repeats, local hours, paid programming, encore blocks and a few specials.
genre	str	E.g. Drama, News, Sports.
program_type	str	series, reality, documentary, news, talk, game show, sports, special, local, paid programming.
broadcast_date	date	The broadcast day the telecast belongs to (days run 06:00–05:59).
daypart	str	Early Morning, Daytime, Early Fringe, Prime Access, Prime, Late Night, Overnight.
start_ts, end_ts	datetime	Air time, US Eastern.
duration_min	int	Length in minutes.
is_live, is_repeat	bool	Live event / repeat airing.

viewings: one row per viewing session

A session is one person (or one household's TV set) on one network without interruption.

COLUMN	TYPE	MEANING
viewing_id	int	Unique session.
respondent_id	int	Household number + two digits: ...00 household (H), ...01, ...02 people, ...81, ...82 guests.
resp_type	str	H, R, L or S.
guest_attr_ids	list[int]	Only for guests: their sex and age-band codes.
network_code	str	Network watched (tuned, for H rows).
start_ts, end_ts	datetime	Session start and end, US Eastern.
playback	str	Live (this file has live viewing only).
weight	int	The respondent's weight for the broadcast day the session starts in (for H: the household's; for S: the household member they watch with).

daily_weights: one row per respondent per in-tab broadcast day

Everyone who was in the panel and measured that broadcast day (**in-tab**), whether they watched these networks or not. A respondent missing on a day was not in-tab: not yet in the panel, gone, or the meter failed. Viewing between 00:00 and 05:59 belongs to the previous broadcast day, so use its weight.

COLUMN	TYPE	MEANING
respondent_id	int	As above (H, R and L).
resp_type	str	H, R or L.
broadcast_date	date	Broadcast day (06:00 that date to 05:59 the next).
weight	int	US households (H) or persons (R, L) represented that day.

`viewer_attributes` and `attributes`: who they are

Characteristics are stored as *codes*, the way Nielsen delivers them. `viewer_attributes` has one row per respondent and code; `attributes` says what each code means (`202` = Female, `335` = Age 35–49, ...). Household characteristics (size, income, children, TV reception, region, county size) sit on the household record (`...00`); person characteristics (sex, age, race, education, employment, household role) on each person. Every record also carries its type code: 104 household, 101 member (R), 102 long-term visitor (L). Guests are not in this file; their sex and age-band codes are in `viewings.guest_attr_ids`.

COLUMN	TYPE	MEANING
`respondent_id`	`int`	Household record or person (H, R, L).
`attr_id`	`int`	Code; look it up in `attributes`.
`value`	`float`	Only for numeric codes: age in years (code 300); empty otherwise.

COLUMN	TYPE	MEANING
`attr_id`	`int`	Code (`attributes` table).
`attr_group`	`str`	E.g. `Sex`, `Age band`, `Household income`, `TV reception`.
`attr_name`	`str`	E.g. `Female`, `35-49`, `$50k-$75k`.
`value_type`	`str`	`flag` (the code itself is the answer) or `numeric` (read `value`).
`applies_to`	`str`	Which record types can carry the code (`H`, `R`, `L`, `S`).

Audiences: who a number is about

TV is bought and sold for a **target audience** (a demographic), and every rating, reach or audience number is *for* one of them. The labels are short codes:

- **HH**: TV **households**. A household counts as watching when its set is tuned to the network (the `H` rows);
- **P2+**: **persons** aged 2 and over, i.e. everyone the panel measures. "P" = persons of both sexes;
- **A18–34**, **A18–49**: **adults** aged 18–34 and 18–49, the classic advertiser targets for younger audiences;
- **M** and **W** (e.g. M18–34, W55+): **men** only or **women** only in that age range.

The same telecast has a different rating for each audience: a rating is the share of *that* audience who watched (Lecture 2: one rating point = 1% of the target audience). An advertiser who sells to young men cares about the M18–34 rating, not the P2+ one. Find a person's age in `viewer_attributes` (code 300 for years, or the age-band codes) and their sex (codes 201/202).

> **WATCH OUT**
>
> **Rules for this lab** (so everyone gets the same numbers):
>
> - a household or person has *watched* a telecast if they watched at least 1 minute of it;
> - **persons** are R and L only; S guests are left out of every person number, because a guest id does not identify a person;
> - a telecast's **rating** = sum of the weights of those who watched it ÷ sum of the weights of everyone in-tab that broadcast day (in the same group: households, P2+, A18–34, ...) $\times 100$;

- for **reach** over several days, give each person one weight for the month: their *average* weight over the broadcast days they were in-tab. Reach = sum of these weights over people who watched at least once ÷ sum over everyone in the group $\times 100$. (The industry uses one fixed weight per person for the whole period; the average is our stand-in for it.)

3 Tasks

Task 1 Get to know the data — 6 PTS

Answer each question with numbers and one or two sentences.

1. **Panel and universe.** How many telecasts, distinct programs and networks are in `programs`? How many households and persons were in-tab at least once in October (use `daily_weights`)? Why does `viewer_attributes` list a few more? On an average day, how many are in-tab, and how many US households and persons do their weights add up to?
2. **When?** Plot the projected **average audience** (millions of persons watching these ten networks per minute) by hour of day, separately for weekdays and weekends. Which hour is the peak? How does the weekend differ?
3. **How long?** Plot the distribution of person session length, weighting each session by its weight. Report the weighted mean and median. Why are they different, and which one would you put in a client deck?

> **HINT**
> A session from 20:50 to 21:20 adds 10 minutes to hour 20 and 20 minutes to hour 21. Split sessions at hour boundaries (or expand them into minutes) and multiply by the weight before summing. Average audience in an hour = weighted minutes in that hour ÷ 60 ÷ number of such days. Count weekdays and weekends by broadcast day: Friday 01:00 still belongs to Thursday's broadcast day.

Task 2 Program reach curve — 2 PTS

Take all October telecasts of *CBS Evening News* (weekdays on CBS).

1. For each telecast compute the **household rating**, the **P2+ rating** and the **projected audience** in millions of persons.
2. In air order, compute the **cumulative reach** after each telecast for **P2+**, **A18–34** and **A18–49**, and plot the three curves.
3. Why does cumulative reach keep growing while each telecast's rating stays roughly flat? Which audience does the program reach best, and does that match what you know about who watches network evening news?

Task 3 Program frequency: how loyal is the audience? — 2 PTS

For *The Young and the Restless* (a daily soap on CBS), count for every person who watched how many of its October telecasts they saw: that is the program's **frequency**, the same idea as ad frequency in Lecture 1. Use each person's monthly weight from the rules.

1. Report the projected number of viewers (millions), and the weighted average and median frequency.
2. Plot the **frequency distribution** in projected viewers: how many millions watched 1, 2, 3, … telecasts.
3. What share of all projected views comes from the most loyal viewers? Define them as everyone whose frequency is *at or above* the weighted 80th percentile of frequency (whole frequency levels, no splitting). Is the average frequency a fair summary ("the average can lie")?

Task 4 Networks by age and gender — 2 PTS

For each network compute projected viewing minutes (minutes × weight) for six audiences: **M18–34**, **W18–34**, **M35–54**, **W35–54**, **M55+** and **W55+**. List the top 5 networks for each audience.

- Which network has the youngest audience and which the oldest? Measure it as the share of each network's projected minutes coming from 18–34 viewers.
- Compute the same share once more *without* weights (raw panel minutes). Why is it different? What does it tell you about who is over- or under-represented in the panel, and what would go wrong if you reported raw panel numbers to a client?
- **Bonus (+1 point):** one or two sentences: if you sold ads on the oldest network, how would you pitch it to an advertiser who wants young adults?

4 What to submit

Submit a **link to your GitHub repository**. The repository must contain:

- `lab1.ipynb`: your notebook, with all outputs (plots and printed numbers) saved. It must run top to bottom on Colab or Kaggle without errors;
- optionally `lab1_helpers.py` with helper functions, imported by the notebook;
- a short `README.md` with your name and group.

Do **not** commit the data files: the setup cell downloads the course dataset from Kaggle with `kagglehub`, which works on both Colab and Kaggle. Make the repository public, or share it with the TA before the deadline. Commits after the deadline are not graded.

Kaggle dataset: `shkarupylomaxim/adtech-lab1-tv-panel`

Notebook structure

Your notebook must follow this skeleton: **the same markdown headings, in the same order, and the same printed labels**. That is how your work is found and graded. Cells are shown in Jupytext "percent" format: `# %% [markdown]` starts a markdown cell, `# %%` a code cell. Replace every `...` with your own code. Add as many cells as you like *between* the headings, and write each written answer in the markdown cell *Answer* under its heading.

```
# %% [markdown]
# # Lab 1 — Intro to Behavioral TV Data
# Name: ... | Group: ...

```

```python
# %% [markdown]
# ## Setup
# %%
import kagglehub
import pandas as pd
import matplotlib.pyplot as plt
DATA = kagglehub.dataset_download("shkarupylomaxim/adtech-lab1-tv-panel/versions/2")
       # Colab and Kaggle
programs = pd.read_parquet(f"{DATA}/programs.parquet")
viewings = pd.read_parquet(f"{DATA}/viewings.parquet")
daily_weights = pd.read_parquet(f"{DATA}/daily_weights.parquet")
viewer_attributes = pd.read_parquet(f"{DATA}/viewer_attributes.parquet")
attributes = pd.read_parquet(f"{DATA}/attributes.parquet")

# %% [markdown]
# ## Task 1.1 — Panel and universe
# %%
print("Telecasts:", ...)
print("Programs:", ...)
print("Networks:", ...)
print("Households in panel:", ...)
print("Persons in panel:", ...)
print("Households in-tab, average day:", ...)
print("Persons in-tab, average day:", ...)
print("US TV households represented, average day (M):", ...)
print("US persons represented, average day (M):", ...)

# %% [markdown]
# ## Task 1.2 — Average audience by hour of day
# %%
...   # plot: projected average audience (M) by clock hour, weekday vs weekend
print("Peak hour, weekdays:", ...)
print("Peak hour, weekends:", ...)
# %% [markdown]
# **Answer:** how the weekend differs.

# %% [markdown]
# ## Task 1.3 — Session length
# %%
...   # plot: weighted distribution of person session length
print("Session length, weighted mean (min):", ...)
print("Session length, weighted median (min):", ...)
# %% [markdown]
# **Answer:** why mean and median differ; which one you would show a client.

# %% [markdown]
# ## Task 2 — Program reach curve
# %%
...   # table: household rating, P2+ rating, projected audience (M) per telecast
...   # plot: cumulative reach P2+, A18-34, A18-49
print("Telecasts of the program:", ...)
print("Average household rating (%):", ...)
print("Average P2+ rating (%):", ...)
print("Average projected audience (M):", ...)
```

```
print("Final reach P2+ (%):", ...)
print("Final reach A18-34 (%):", ...)
print("Final reach A18-49 (%):", ...)
# %% [markdown]
# **Answer:** reach vs rating; which audience is reached best, and why.

# %% [markdown]
# ## Task 3 — Program frequency
# %%
...   # plot: projected viewers (M) by number of telecasts watched
print("Projected viewers (M):", ...)
print("Average frequency (weighted):", ...)
print("Median frequency (weighted):", ...)
print("Share of views from the most loyal viewers:", ...)
# %% [markdown]
# **Answer:** is the average frequency a fair summary?

# %% [markdown]
# ## Task 4 — Networks by age and gender
# %%
...   # six top-5 tables (projected minutes): M18-34, W18-34, M35-54, W35-54, M55+, W55+
...   # table: 18-34 share of minutes per network, weighted and raw
print("Youngest network:", ...)
print("Oldest network:", ...)
# %% [markdown]
# **Answer:** weighted vs raw.
# **Bonus answer:** the sales pitch.
```

> NOTE
>
> Print numbers as plain values (no units in the value itself); percentages as 0–100 with one decimal, shares as 0–1 with three decimals, audiences in millions with two decimals.

5 Grading

CRITERION	POINTS
Task 1.0: importing the data and setting up the workflow (repository, notebook runs top to bottom, codes joined to their names)	3
Task 1: panel and universe, audience by hour, session length	3
Task 2: household and person ratings, projected audiences, three reach curves, interpretation	2
Task 3: weighted frequency figures and distribution, interpretation	2
Task 4: top networks for six audiences, youngest and oldest, weighted vs raw	2
Total	**12 / 12**

> NOTE
>
> **Bonus:** +1 point for the sales pitch in Task 4 (the lab total can reach 13 of 12). Individual work. Late work: -2 points per started week, accepted until December 13.

FURTHER READING

Lecture 1: reach, frequency and “the average can lie” · Lecture 2: a rating point is 1% of the target audience · Nielsen, How are TV audiences measured?